

Elementary Matrices: Construction and Algebraic Properties

High-Level Overview

Up to this point, we have used elementary row and column operations as tools to simplify matrices and compute their rank. But what if we could represent these operations as **independent matrices** themselves?

An **elementary matrix** is a special matrix obtained by applying a single elementary transformation directly to an identity matrix (I).

Why is this important in Engineering and Computer Science?

- **Matrix Factorization:** Representing operations as matrices allows computers to factor complex matrices into simpler components (like LU decomposition), which speeds up the numerical solution of massive systems of equations.
- **Invertibility:** Every elementary matrix is square and invertible, meaning you can "undo" any elementary transformation by multiplying by its inverse elementary matrix.

1. Definition and Construction of Elementary Matrices

An elementary matrix E of order $n \times n$ that performs a specified elementary operation is constructed by performing that exact same operation on an identity matrix of equal order.

Examples of Elementary Matrices

Applying the three types of elementary operations to a 3×3 identity matrix yields:

1. Row/Column Interchange:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

2. Scalar Multiplication (k times a row):

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3. Row/Column Addition (k times one row added to another):

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Pre-Multiplication vs. Post-Multiplication

The location of the elementary matrix relative to matrix A determines whether the operation affects the **rows** or the **columns**.

Theorem:

- **Pre-multiplying (Left Multiplication - EA):** Performs an **elementary row operation** on A .
- **Post-multiplying (Right Multiplication - AE):** Performs an **elementary column operation** on A .

Worked Example: Constructing and Applying an Elementary Matrix

Let's look at how left and right multiplication alter a matrix in practice.

Given matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

And an elementary matrix E that swaps the first and second rows (derived by swapping rows 1 and 2 of the 3×3 identity matrix):

$$E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Case A: Pre-Multiplication (EA) - Row Transformation

Multiplying from the left applies the operation to the rows of A :

$$EA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$$

- **Result:** The first and second rows of A are successfully interchanged.

Case B: Post-Multiplication (AE) - Column Transformation

Multiplying from the right applies the operation to the columns of A :

$$AE = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 5 & 4 & 6 \\ 8 & 7 & 9 \end{bmatrix}$$

- **Result:** The first and second columns of A are successfully interchanged.

Summary Checklist for Rapid Revision

- ☐ **Definition:** An elementary matrix is built by applying a single elementary operation to an **identity matrix** (I).
- ☐ **Left Multiplication** (EA): Performs elementary **row** operations.
- ☐ **Right Multiplication** (AE): Performs elementary **column** operations.
- ☐ **Invertibility:** Every elementary matrix has a corresponding inverse elementary matrix that reverses the transformation.